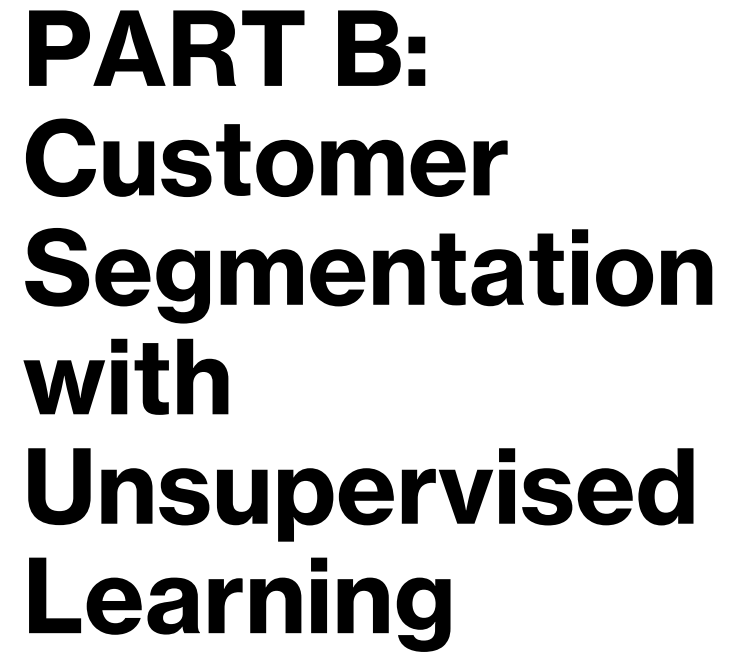




Admin: P2423500

Class: DAAA/FT/1B/0

Project Overview

Background:

We collected the information of **200 customers** including their **Gender, Age, Income and How much they spend**. We want to group the customers into **clusters** so that we can **cater needs for different customers**.

Phases of this project

- 1. Data Collection and Preprocessing**
- 2. Exploratory data analysis (EDA)**
- 3. Implementing Multiple Clustering Models**
- 4. Clustering Model Comparison and Final Insights**
- 5. Business Insights and Final Conclusions**
- 6. Answering the business questions with visualizations**

Phase 1: Data Collection and preprocessing

- First we check for the data information using data.info
- Check for missing values
- Identify what columns we're working with.

```
# Load the dataset
df = pd.read_csv("CA2-Customer-Data.csv")

# Display first few rows
print("First 5 rows of the dataset:")
print(df.head())

# Check dataset information
print("\nDataset Information:")
print(df.info())
```

First 5 rows of the dataset:

	CustomerID	Gender	Age	Income (k\$)	How Much They Spend
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   object
2   Age                  200 non-null   int64
3   Income (k$)          200 non-null   int64
4   How Much They Spend  200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None
```

```
# Summary statistics
print("\nSummary Statistics:")
print(df.describe())

# Check for missing values
print("\nMissing values per column:")
print(df.isnull().sum())

# Check for duplicates
duplicate_count = df.duplicated().sum()
print(f"\n Duplicate Rows Found: {duplicate_count}")
```

Summary Statistics:

	CustomerID	Age	Income (k\$)	How Much They Spend
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

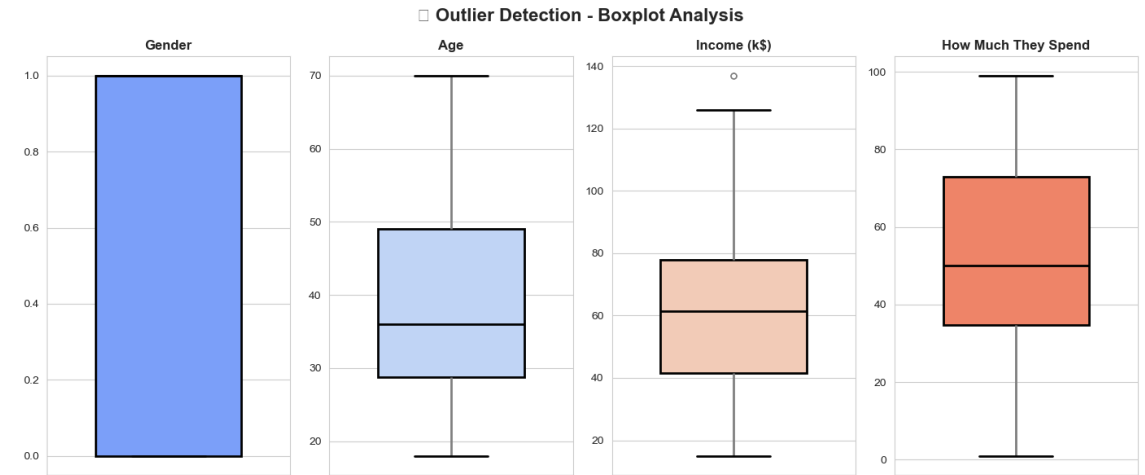
Missing values per column:

```
CustomerID    0
Gender        0
Age           0
Income (k$)   0
How Much They Spend  0
dtype: int64
```

Duplicate Rows Found: 0

Phase 1: Data Collection and preprocessing

- Next we look for outliers using Z Score and IQR method.
- Then we handle them using winsorization by capping the outliers at the top and bottom 1% values.
- Then we proceed to feature engineering by creating new features that may help make clustering easier.
- We feature engineered: **spending efficiency** and **age-income ratio**



```
# Winsorization: Capping extreme values
df["Income (k$)"] = winsorize(df["Income (k$)"], limits=[0.01, 0.01]) # Cap top & bottom 1%
df["How Much They Spend"] = winsorize(df["How Much They Spend"], limits=[0.01, 0.01])

# Feature Engineering - Creating new derived features
df["Spending Efficiency"] = df["How Much They Spend"] / df["Income (k$)"]
df["Age-Income Ratio"] = df["Income (k$)"] / df["Age"]

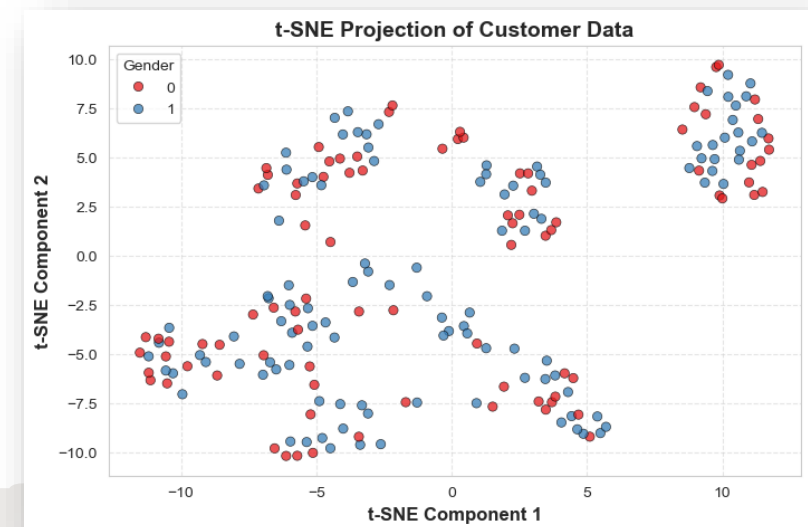
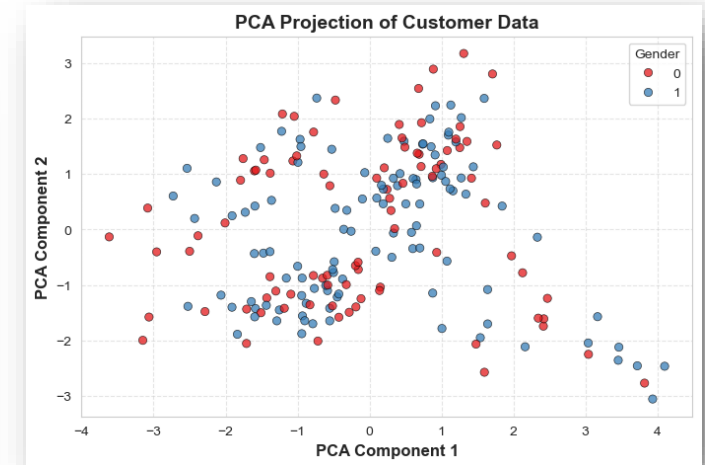
# Display dataset after processing
print("\n Updated Dataset After Feature Engineering:")
df.head()
```

Phase 1: Data Collection and preprocessing

- Next we do feature scaling and dimensionality reduction.
- Feature scaling is a standardization technique used to make all features contribute equally to the clustering process by bringing them to a common scale.
- Here I used Z score standardization which standardizes features to have **Mean: 0 and S.D: 1**.
- After, I did dimensionality reduction with PCA technique and the t-SNE Non linear dimensionality reduction technique from further research.
- What PCA does is to reduce high dimensionality data into 2 components (PCA1 and PCA2) while preserving maximum variance. Here we used `n_component = 2` for 2D visualization.
- What t_SNE Non Linear dimensionality reduction is, is that it reduces dimensions while preserving their relationships (Distance and how similar they are). This is a more complex technique that makes **visualizing the clusters easier**. Only cons being that it is slow on larger datasets.

```
PCA Reduced Dataset:
  PCA1    PCA2  Gender
0  2.127096 -0.779575    0
1  3.821184 -2.766737    0
2  0.879225  0.932429    1
3  3.721171 -2.455149    1
4  2.332363 -0.138528    1

t-SNE Transformed Dataset:
  tSNE1    tSNE2  Gender
0  1.515202 -7.657561    0
1  5.101877 -9.184182    0
2 -2.617802 -9.565855    1
3  4.862892 -9.036869    1
4  0.918088 -7.482295    1
```

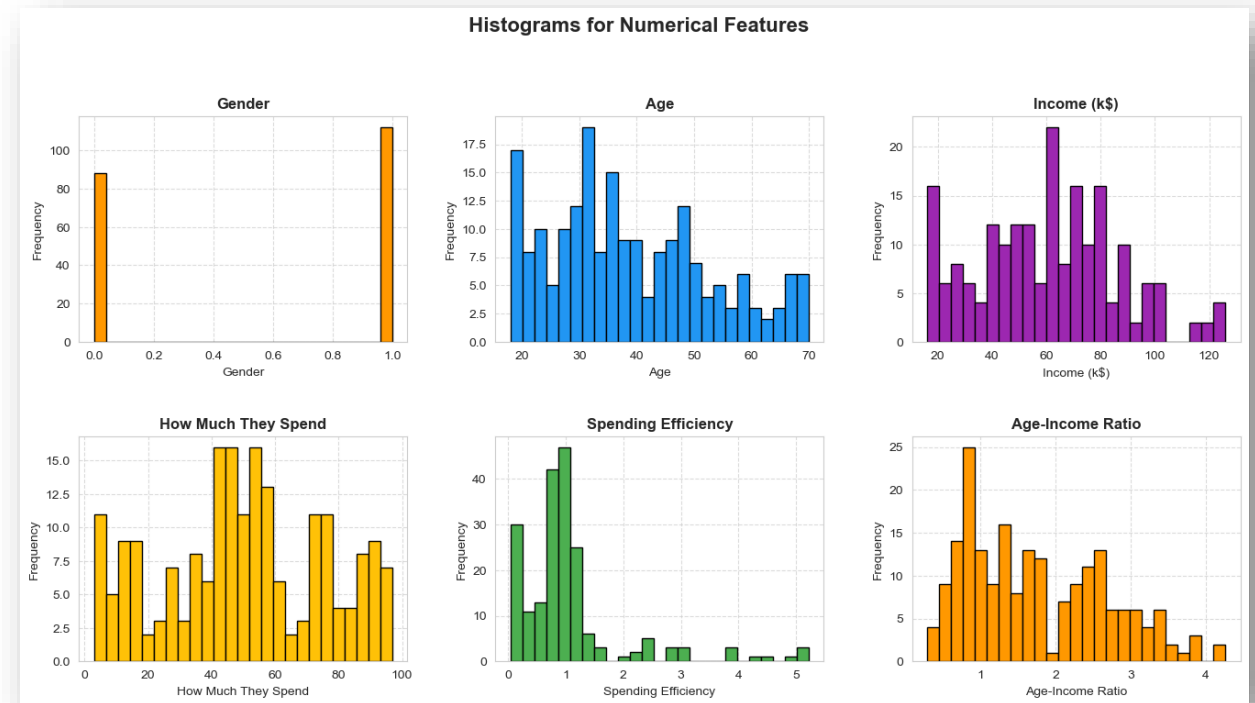


Phase 2: Exploratory Data Analysis

- Here we plotted a histogram for the numerical features to see their distribution of data.

We can see that:

- Majority of the customers are between ages: 20-50, with fewer customers above 60 yrs of age.
- Income sees two peaked incomes, with peaks at 60k and 80k per year, with a small few high income customers.
- Most customers spend between 40 to 60 units. While some are low spenders, the rest are high spenders too.
- Spending efficiency extremely right skewed, indicating low income, high spending behaviour.
- age-income ratio right skewed as well, with most values under 2.5, indicating younger customers generally have higher income, with the tail being the older customers.

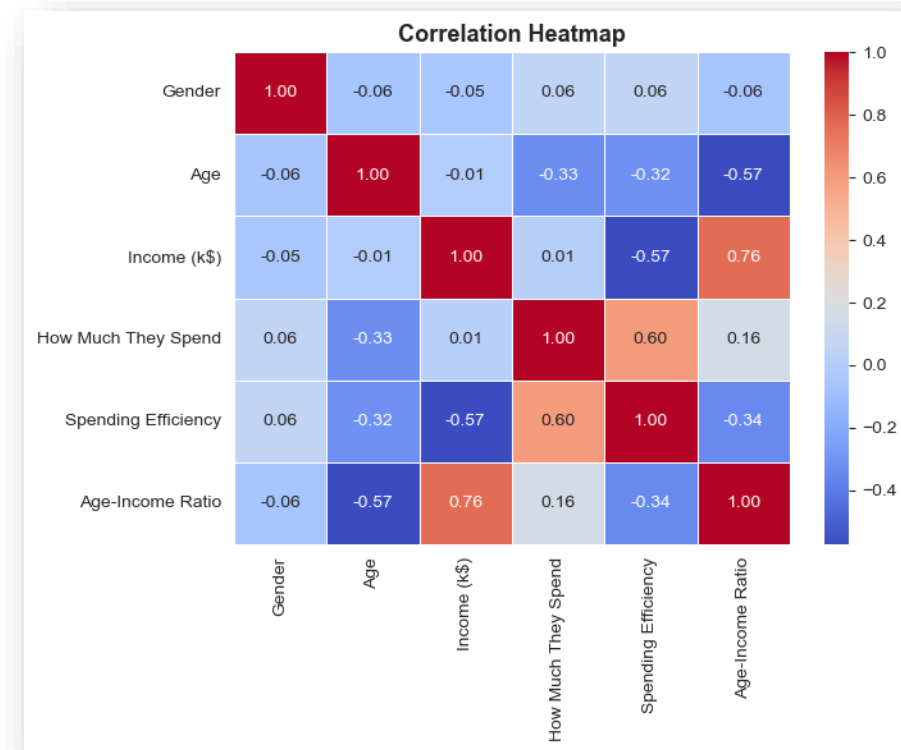


Phase 2: Exploratory Data Analysis

- Here we plotted a correlation heatmap to spot for relationships between one another.

Here we can see that:

- Gender has no significant impact on income, spending or efficiency.
- Age and spending has a rather low correlation of -0.33, meaning younger customers tend to spend more, but relationship is rather weak.
- High correlation factors such as income, spending efficiency, age income ratio can contribute well to the cluster separations.

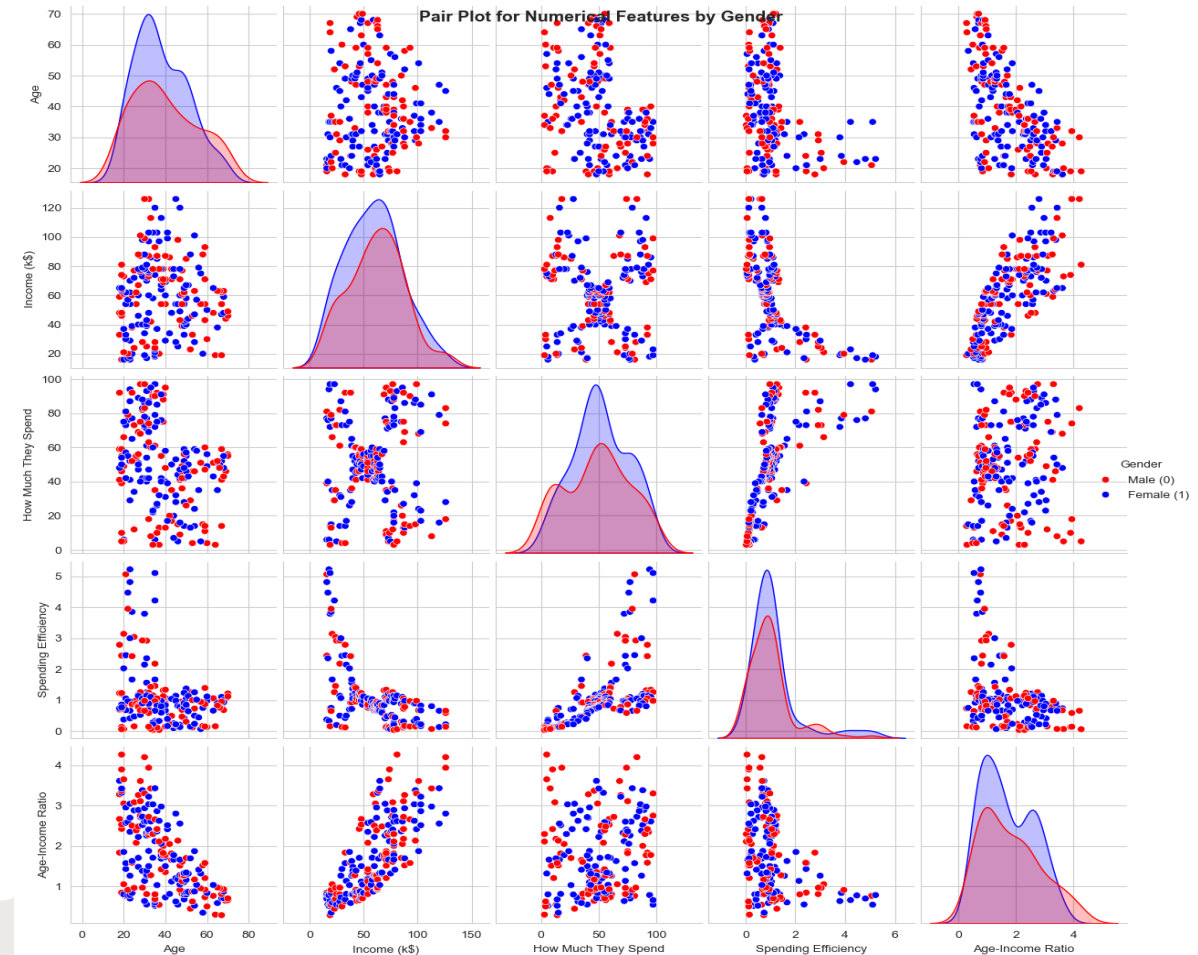


Phase 2: Exploratory Data Analysis

- Here we plotted a pair plot, for male and female.

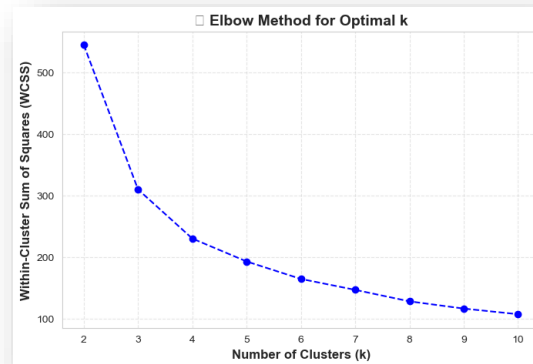
From this we can see that:

- There are more customers aged 30-50, with the males being a bit more older in general.
- Income is normally distributed, with the males earning a slightly higher income than the females.
- Customers usually spend around 20k-80k with a small group spending above 100k, with females having a wider range of spending habits.
- Younger customers (20-40) show a wide range of incomes, suggesting a diverse customer base.

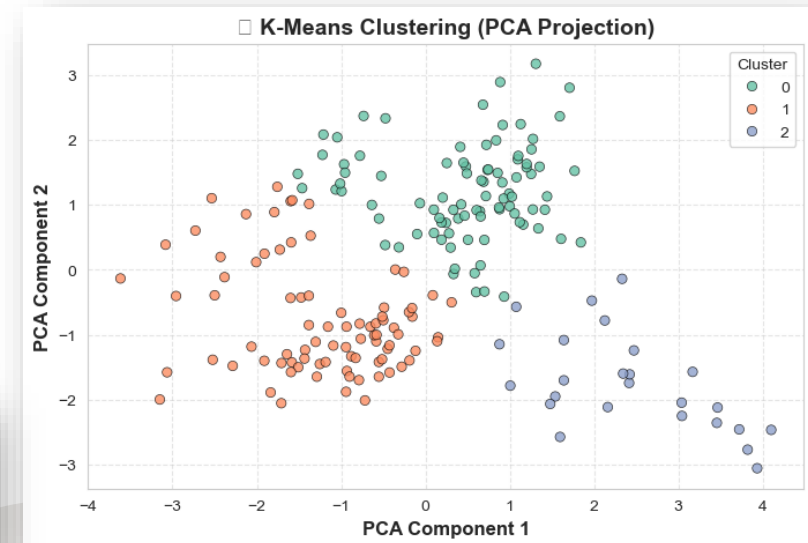
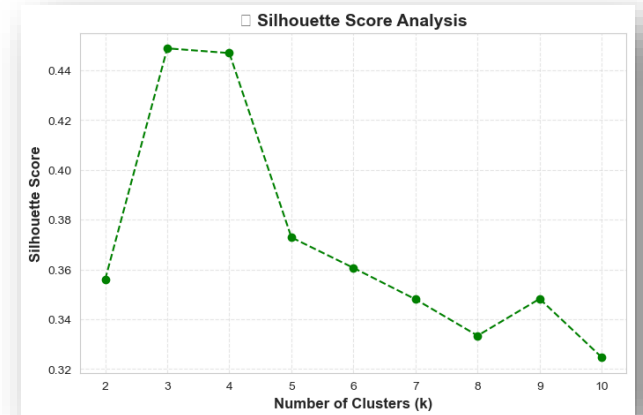


Phase 3: Implementing multiple clustering models.

- Here we first implement K-Means Clustering. Which is a centroid based clustering technique.
- We first need to use the elbow method & Silhouette score technique to find the optimal K value.
- K value is the number of clusters to create.
- How it works is that it chooses the k value and places k centroids randomly, and assign each point to the nearest centroid, then move the centroids to the center of their assigned points.
- From elbow method and the silhouette score, we can see that the optimal k value we have is 3.
- We then calculate the performance of this clustering technique with Silhouette score, davies-bouldin score and calinski-harabasz score.

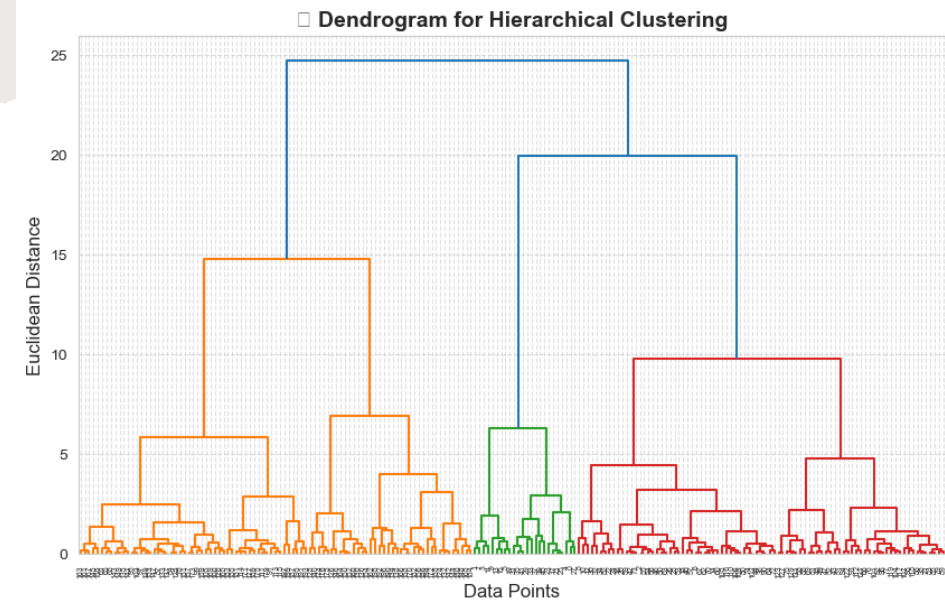


Silhouette Score (Higher is Better): 0.4945
Davies-Bouldin Score (Lower is Better): 0.7064
Calinski-Harabasz Score (Higher is Better): 209.4535

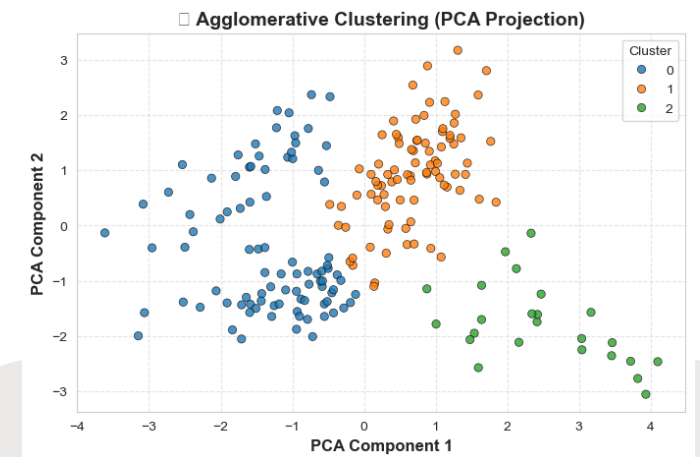


Phase 3: Implementing multiple clustering models.

- Here we implement another clustering model called Agglomerative (hierarchical clustering).
- We first need the K value, as seen from the dendrogram, which we can see is $k = 3$.
- We then use this for clustering and assess the results using the same scoring system again.

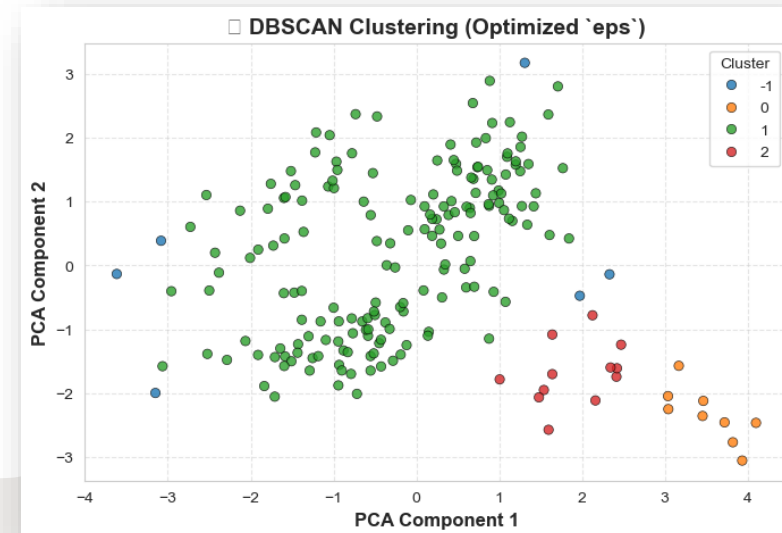
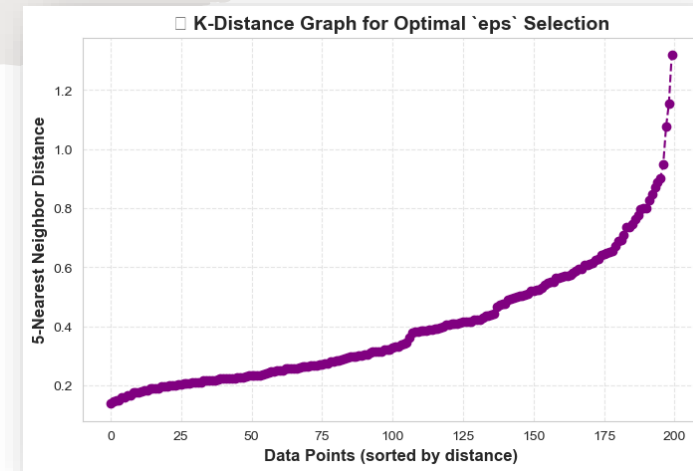


Silhouette Score (Higher is Better): 0.4287
Davies-Bouldin Score (Lower is Better): 0.8170
Calinski-Harabasz Score (Higher is Better): 160.9326



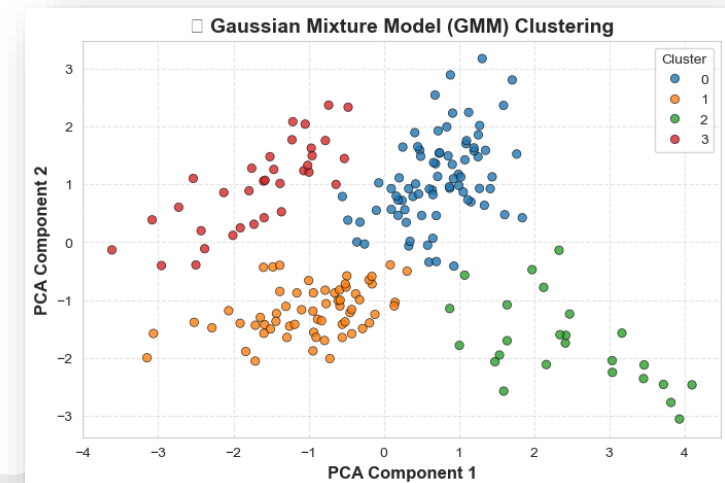
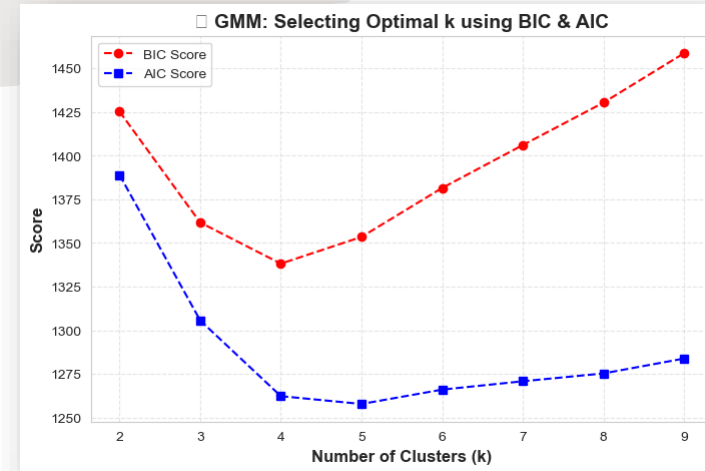
Phase 3: Implementing multiple clustering models.

- Here we implement the DBSCAN clustering model.
- DBSCAN is a density-based clustering algorithm that groups them based on how close they're packed together.
- First it picks a random, unvisited point, and check its neighbouring distance within distance “eps” (epsilon), and if the point has at least min_samples neighbours, it forms a cluster, or else its marked as noise.
- To find the optimal eps, we check the K-Distance Graph. We can identify the best eps parameter by checking where the graph starts to bend sharply, which we can see its at around 0.65 to 0.7, hence we'll use 0.65 as the best eps parameter.
- We then use this to do clustering, and save its scores for the 3 scoring systems we use earlier.



Phase 3: Implementing multiple clustering models.

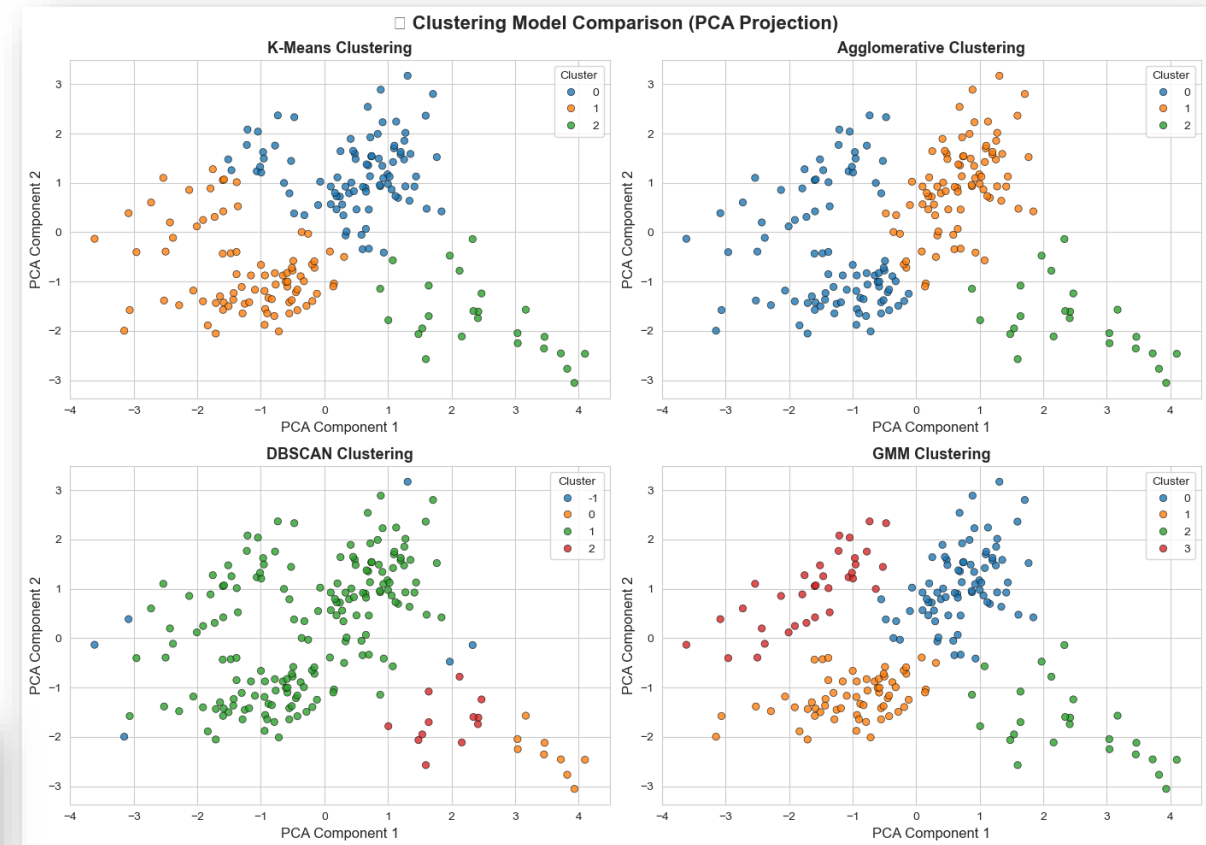
- For our last model, we will implement Gaussian Mixture Model Clustering. A model that uses probability and assumes the data is generated from a mixture of gaussian distribution.
- To find the best K, we will need to find the optimal K using AIC and BIC plot.
- From the plot, we can see that the optimal K is 4, because after k=4, the scores increase, indicating overfitting.
- Both the curves support K = 4, which means it's the optimal K.
- We then use the K value to cluster, and store its score assessed with those 3 scoring systems we used earlier.



Phase 4: Clustering Model Comparison & Final Insights.

- Here we compile all the clusterings, as well as their respective scores to figure out which clustering does the best, according to their scores.
- They will be scored by the silhouette score, Davies-Bouldin Score and Calinski-Harabasz Score.
- From the results below, we can see that the GMM is the best clustering model, scoring the best across all 3 metrics, with K-Means being the second best clustering model.

	Model	Silhouette Score	Davies-Bouldin Score	Calinski-Harabasz Score
0	K-Means	0.494521	0.706431	209.453514
1	Agglomerative	0.428737	0.816974	160.932565
2	DBSCAN	0.357510	0.709719	50.864360
3	GMM	0.505965	0.687374	223.951882



Phase 5: Business Insights and Final Conclusions

- Here, we analyze the cluster that GMM has made, to fulfill the given task to **identify characteristics of each customer cluster** and **which group of customers to pay more attention to**.
- We can see the clustering at the right, from that, we can determine 4 different clusters mainly:
 1. **Older Moderate Spenders**
 2. **Younger High income, high spending**
 3. **Younger, low income, high spending**
 4. **High income, low spending.**

```
# Compute the average characteristics for each customer cluster
gmm_cluster_profiles = df.copy()
gmm_cluster_profiles["GMM_Cluster"] = df_pca["GMM_Cluster"]
cluster_summary = gmm_cluster_profiles.groupby("GMM_Cluster").mean()

cluster_summary
```

	Gender	Age	Income (k\$)	How Much They Spend	Spending Efficiency	Age-Income Ratio
GMM_Cluster						
0	0.602564	49.948718	46.794872	40.923077	0.871958	0.978798
1	0.555556	28.904762	76.142857	69.714286	0.931917	2.680953
2	0.600000	25.480000	25.560000	75.320000	3.199348	1.044202
3	0.441176	41.647059	88.411765	16.882353	0.190248	2.304990

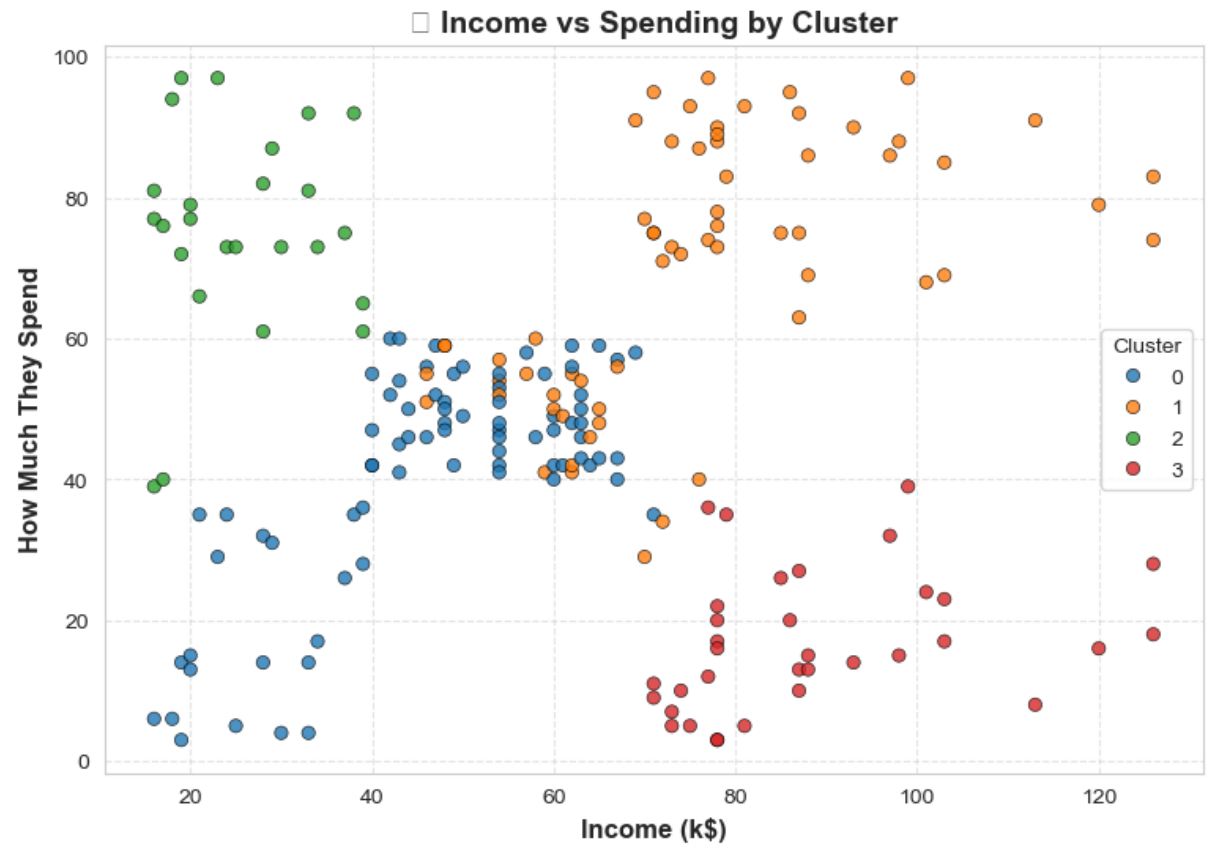
Phase 6: Answering business questions with visualizations.

- Now we will plot graphs and analyze with visualizations to answer business questions that the businesses would have, and answering them, as well as identifying the cluster to pay more attention to.
- First we plot a bar graph to check the size of each cluster. From this we can see the cluster 0 (**older moderate spenders**) are most of the customers.



Phase 6: Answering business questions with visualizations.

- Next, we plot a scatter plot to visualize the income vs spending by cluster.
- From the plot we can tell that:
 1. Cluster 0, (Mid income, mid spending) have moderate incomes, spending is also moderate and consistent, indicating these are financially stable, cost-conscious customers
 2. Cluster 1, (high income, high spending), have higher incomes, but also spend more, meaning we should focus on offering these customers premium services.
 3. Cluster 2, (low income, high spending), have lower incomes, but spend quite a lot reaching 90k, businesses can offer installment plans or loyalty discounts, etc.
 4. Cluster 3, (high income, low spending), have higher incomes above 80k, but don't spend as much, businesses should offer more targeted promotions.



Phase 6: Answering business questions with visualizations.

- To conclude which cluster needs the most attention, we can organize the customer category to how much at risk the businesses are losing them, judging by their average spending against income.
- Below we can see that, the first cluster needs the most attention as they are at risk, however they are usually quite stable and do not see much growth, and are already a huge portion of the customer-base.

```
# Use cluster_summary directly (no cluster ranking needed)
high_spending_threshold = cluster_summary["How Much They Spend"].mean()
high_income_threshold = cluster_summary["Income (k$)"].mean()

# Function to assign categories based on spending and income
def categorize_cluster(row):
    if row["How Much They Spend"] >= high_spending_threshold and row["Income (k$)"] >= high_income_threshold:
        return "High-Value Customers"
    elif row["How Much They Spend"] >= high_spending_threshold and row["Income (k$)"] < high_income_threshold:
        return "Potential Growth Customers"
    elif row["How Much They Spend"] < high_spending_threshold and row["Income (k$)"] >= high_income_threshold:
        return "Budget-Conscious Customers"
    else:
        return "At-Risk Customers"

# Apply categorization to cluster_summary
cluster_summary["Customer Category"] = cluster_summary.apply(categorize_cluster, axis=1)

cluster_summary
```

	Gender	Age	Income (k\$)	How Much They Spend	Spending Efficiency	Age-Income Ratio	Customer Category
GMM_Cluster							
0	0.602564	49.948718	46.794872	40.923077	0.871958	0.978798	At-Risk Customers
1	0.555556	28.904762	76.142857	69.714286	0.931917	2.680953	High-Value Customers
2	0.600000	25.480000	25.560000	75.320000	3.199348	1.044202	Potential Growth Customers
3	0.441176	41.647059	88.411765	16.882353	0.190248	2.304990	Budget-Conscious Customers

Final Conclusion

- From that, we can conclude to answer which cluster needs the most attention:
- **Cluster 2 requires the most attention because they are spending the most, relative to their income, businesses can offer financial planning services, and loyalty discounts to keep them.**
- While for other clusters:
- Cluster 3 are the ideal cluster to offer premium services to.
- Cluster 1 are the ideal cluster to offer luxury products and high end services due to their high spending
- Cluster 0 are the ideal clusters to offer financial security plans, and family friendly services as they are relatively older.